

Water Dam Monitoring with Weather Integration

Alexandre Cunha^[0009–0005–9408–5221], Artur Oliveria^[1111–2222–3333–4444],
Diogo Fernandes^[2222–3333–4444–5555], Diogo Silva^[2222–3333–4444–5555], Félix
Martins^[2222–3333–4444–5555], Gonalo Santos^[2222–3333–4444–5555], and Pedro
Barbeira^[2222–3333–4444–5555]

Department of Informatics Engineering,
Faculty of Engineering, University of Porto, Porto, PT
{up202400086, up202108663, up202108752, up202104794,
up202108837, up202108839, up201303693}@fe.up.pt

Abstract. The abstract should briefly summarize the contents of the paper in 150–250 words.

Keywords: First keyword · Second keyword · Another keyword.

1 Introduction

The management of hydroelectric dams represents a critical engineering challenge, as these infrastructures are vital for both sustainable water supply and electricity generation. This project is motivated by the inherent risks associated with dam operations: the danger of overflow during periods of heavy rainfall, which presents significant flooding and safety risks, and the threat of drought during dry seasons, which leads to reduced energy production. Currently, monitoring methods can often be inefficient or outdated. Consequently, there is a clear need for a smart, autonomous, and remote monitoring system to unify and optimize these industrial processes.

The primary objective of this project is to develop a Cyber-Physical System (CPS) and Internet of Things (IoT) framework for dam monitoring and control. The proposed solution collects real-time data on variables such as water level, flow rates and rainfall intensity, supporting compliance with the legally required safety control and observation measures defined for each dam class and associated damage potential, in accordance with the Portuguese dam safety framework established by Decreto-Lei n.^o 21/2018, de 28 de maro[1], which regulates the safety of large and small dams throughout the phases of design, construction, first impoundment and operation. While the legislation does not prescribe a universal operating range in terms of reservoir percentage, this work adopts a conservative operational strategy in which the CPS helps to keep the reservoir within project-defined safe limits (for example, between 20% and 90% of storage capacity). This is achieved through a distributed architecture utilizing Arduino Nano ESP32 units for local data acquisition and a Raspberry Pi gateway running a Node-RED server and MQTT broker. Furthermore, the system supports predictive decisions using weather forecasts and historical data processed via Gradient

Boosting Machine (GBM) models, all visualized through a remote Node-RED dashboard for manual and automatic gate control.

2 Fundamental Knowledge and Related Work

2.1 Review of Existing Systems

2.2 Gap Analysis

3 Proposed Solution

The proposed solution architecture is a multi-layered Cyber-Physical System (CPS) designed for reliability and scalability. It integrates physical sensing, edge computing, and cloud-based data management to create a resilient monitoring loop for hydroelectric dam operations.

3.1 System Design and Architecture

System Architecture and Layers The system is structured according to the 5-layer IoT architecture, ensuring a clear separation between physical acquisition and business logic:

- **Perception Layer (Smart Connection Level):** This layer consists of Arduino Nano ESP32 controllers interfaced with pressure transducers for level monitoring, water flow sensors, and pluviometers.
- **Network Layer:** Communication is established via MQTT over Wi-Fi for the prototype, utilizing a "Topic Strategy" that prioritizes single-input trunks (e.g., `sensors/#`) to ensure all incoming data is captured automatically.
- **Middleware Layer (Cyber Level):** A Raspberry Pi serves as the central gateway, hosting a RabbitMQ message broker for reliable delivery and an InfluxDB time-series database for high-performance storage.
- **Application and Business Layers:** Node-RED provides the core logic engine and Human-Machine Interface (HMI). It processes data, triggers safety alerts if the dam exceeds safe limits (20%–90%), and presents real-time visualizations for manual override capabilities.

CPS 5C Architecture Mapping The system is further analyzed through the lens of the 5C functional architecture for Cyber-Physical Systems, which facilitates a transition from raw physical data to autonomous configuration.

1. **Connection (Data Acquisition):** This initial layer focuses on the seamless acquisition of data from the physical asset. In our system, this involves the integration of pressure transducers, pluviometers, and power output meters, alongside solenoid valve actuators.
2. **Conversion (Data Preprocessing):** Raw data is processed at the edge to become "smart" and structured. Our implementation utilizes Node-RED for data aggregation, where local pluviometer data is combined with external Weather API streams.
3. **Cyber (Data Processing and Analysis):** This layer creates the "Digital Twin" of the dam. Centralized processing is handled by the RabbitMQ message broker, which ensures data persistence and provides a virtual representation of the system state for historical analysis and storage.

4. **Cognition (Awareness and Insight):** This level applies advanced analytics. We employ Gradient Boosting Machine (GBM) models for AI-based inflow/outflow forecasting, which generates high-level awareness of current conditions to support a Decision Support System (DSS).
5. **Configuration (Self-adaptation and Control):** This final level enables the feedback loop. The system translates AI-driven insights into actual commands (e.g., opening a valve by 15%) while maintaining a "Human-in-the-Loop" through the Node-RED dashboard for manual overrides.

Data and Ingestion Strategy The system employs an "Atomic Messaging" philosophy based on a decoupled architecture. Rather than sending complex bundled packets, each physical device acts as separate logical sensors, transmitting independent MQTT messages for each measurement type. This approach ensures the backend remains generic and can scale to new sensor types without structural changes.

Every sensor payload follows a strict JSON "Data Contract" to maintain consistency across the platform:

- **id:** A unique UUID for hardware identification.
- **kind:** The measurement type (e.g., `flow_rate`), which Node-RED maps directly to the InfluxDB measurement name.
- **units:** Saved as metadata (Tags) to differentiate between measurement scales.
- **data:** The raw numerical value used for real-time graphing and historical analysis.
- **time_stamp:** A Unix timestamp used to index data points accurately on the temporal X-axis.

3.2 Database Selection and Logic

InfluxDB was selected as the primary data store due to its native optimization for time-series streams. Unlike relational databases, it provides high-ingest performance for million-point streams and automated lifecycle management through Retention Policies (e.g., `KEEP 30 DAYS`), preventing storage exhaustion. Data ingestion is handled directly within Node-RED to allow for logic-first preprocessing, enabling the system to filter "bad" readings or outliers before they reach the database.

3.3 Implementation Details

The development of the prototype was executed through a hardware-software integration process, prioritizing modularity and real-time responsiveness. The implementation details are categorized as follows:

Hardware Setup and Data Acquisition The physical layer utilizes an Arduino Nano ESP32 units for local data acquisition. These controllers interface with a diverse sensor suite, including water flow sensors for monitoring

inflow and outflow rates, rain gauges (pluviometers) for rainfall intensity, and pressure sensors that determine water levels via hydrostatic pressure. Additionally, power monitoring sensors track turbine voltage and current to calculate energy output. For the prototype, sensor data is simulated and scaled to mirror real-world hydrological conditions.

Communication Infrastructure Data transmission is built on a decoupled "Atomic Messaging" philosophy. Each sensor measurement (e.g., temperature, distance, flow) is sent as an independent MQTT message to its own specific topic (e.g., `sensors/temperature/ws_01`). While the final deployment is designed to scale to LoRaWAN due to its high range, the current prototype utilizes MQTT over Wi-Fi for efficient data transfer to the Raspberry Pi gateway.

Middleware and Backend Logic The Raspberry Pi serves as the central processing hub, running a Dockerized stack consisting of RabbitMQ, InfluxDB, and Node-RED.

- **Message Broker:** RabbitMQ handles the routing of MQTT messages, ensuring reliable and secure delivery through TLS encryption.
- **Data Storage:** InfluxDB was implemented for time-series storage, utilizing "Retention Policies" to automatically manage data lifecycles and prevent storage saturation.
- **Flow Engine:** Node-RED is the primary logic coordinator, responsible for filtering sensor readings and transforming JSON payloads into database-ready formats.

Predictive Analytics Implementation To support proactive dam management, a Gradient Boosting Machine (GBM) model was developed in Python. The model was trained using historical hydrological data from the *Sistema Nacional de Informação de Recursos Hídricos* (SNIRH) and integrated with the OpenWeather API. This allows the system to engineer features based on recent flow values and atmospheric pressure to predict daily inflow/outflow trends.

Human-Machine Interface (HMI) The system provides a comprehensive dashboard built in Node-RED for real-time visualization of all monitored variables. This interface includes automated alerts for overflow or drought conditions and provides manual override buttons for solenoid valve control, ensuring a "Human-in-the-Loop" safety mechanism.

3.4 Implementation Constraints and Design Deviations

During the transition from the theoretical system design to the physical prototype, several constraints necessitated deviations from the original proposed solution. These changes highlight the challenges of hardware availability and the adaptation required for small-scale modeling.

Water Level Sensing: Pressure vs. Ultrasonic Distance The initial design proposed a hydrostatic pressure transducer to measure water levels. However, for the current prototype implementation, an ultrasonic distance sensor was utilized as a primary alternative. While a pressure sensor measures the weight of

the water column from the bottom, the ultrasonic sensor measures the distance from the top of the reservoir to the water surface. This shift required a modification in planing for the reservoir model. The ultrasonic sensor, is cost-effective and easier to implement in a small-scale setup but extremely complicated to apply to a real world scenario.

Flow Control: Manual vs. Automated Actuation The proposed architecture included a solenoid valve as an automated actuator to regulate water flow based on AI-driven commands from the Cognition layer. Due to hardware procurement limitations, the physical actuation is currently performed manually. To maintain the integrity of the Cyber-Physical loop, the system still generates "control recommendations" via the Node-RED dashboard. The operator acts as the physical link in the feedback loop, executing manual gate adjustments in response to the system's predictive alerts for overflow or drought.

3.5 Benefits and Limitations

We expect that the prototype successfully demonstrates the feasibility of a unified IoT monitoring stack, yet the implementation deviations provide valuable insights into system robustness.

The use of an ultrasonic sensor can be a good alternative for small-scale testing on a controlled environment, though it remains sensitive to surface turbulence and even more susceptible to noise from particles in the air compared to hydrostatic methods. Furthermore, while the current manual control of the dam gate is a limitation of the physical prototype, it can validate other aspects of prototype such as the automated and manual decision control and configuration. This ensures that even without direct electronic actuation, the decision support system provides actionable intelligence that enhances operational safety. Future iterations can focus on re-integrating solenoid actuators to achieve the full "Self-Adaptation" envisioned in the 5C CPS architecture.

4 Evaluation and Results

4.1 Testing Methodology

4.2 Case Studies

4.3 Result Analysis

5 Discussion

5.1 Benefits

5.2 Limitations

6 Conclusion and Future Work

6.1 Future Work

Building upon the current prototype, several directions for future research and development have been identified to transition this system toward a real-world deployment:

- **Sensor Fusion and Precision:** Future iterations will involve testing a wider array of sensors to increase the precision of the Decision Support System (DSS). Integrating additional environmental variables will provide the AI models with a more comprehensive dataset, leading to more accurate automated and manual control recommendations.
- **Actuator Integration for Full Automation:** A primary goal is the addition of a physical solenoid valve or similar actuator to replace the current manual intervention. This will enable the system to close the feedback loop entirely, achieving the "Self-Adaptation" level of the 5C architecture where the CPS can autonomously adjust gate positions based on real-time and predictive data.
- **Comparative Level-Sensing Analysis:** We plan to conduct comparative experiments between ultrasonic distance sensors and hydrostatic pressure sensors. By evaluating the performance of each individually and testing a hybrid configuration (using both sensors simultaneously), we aim to determine which solution provides the highest reliability under varying conditions, such as surface turbulence or debris interference.
- **Real-Dam Adaptation:** The final stage of development involves scaling the communication infrastructure from Wi-Fi to LoRaWAN to meet the range requirements of a real dam site. This phase will also include adapting the software logic to handle the massive data throughput and environmental complexities inherent in large-scale hydroelectric operations.

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Disclosure of Interests. It is now necessary to declare any competing interests or to specifically state that the authors have no competing interests. Please place the statement with a bold run-in heading in small font size beneath the (optional) acknowledgments¹, for example: The authors have no competing interests to declare that are relevant to the content of this article. Or: Author A has received research grants from Company W. Author B has received a speaker honorarium from Company X and owns stock in Company Y. Author C is a member of committee Z.

¹ If EquinOCS, our proceedings submission system, is used, then the disclaimer can be provided directly in the system.

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